



Description

Small single turn absolute opto-electronic Converter of angular displacements (absolute encoder) with the fit on the shaft. To compensate for misalignment and ease of installation provides an elastic element.

case Diameter 37.5 mm, the diameter of the through hollow shaft 8 mm resolution up to 2097152 position on the back (up to 21 bits), supply voltage +5 V, serial data interfaces RS422, SSI and BiSS Interface BiSS C. C is the preferred and recommended for use in new projects. Possible results of the incremental signal CH (1Vpp) 1024 period/revolution for positioning at high speeds of rotation of the shaft.

is Used as the feedback sensor of the motor, and a displacement sensor or a speed to the wood and metal equipment, bench and test equipment for the needs of automation in the food industry and in many other areas.

» [Link to product page](#)

Scale	Circular glass scale with absolute track
Design feature	Hollow shaft
Weight (without cable)	~0.1 kg
Shaft diameter	8 mm
Rotor breakaway torque	$\leq 0,005 \text{ Nm}$
Rotor moment of inertia	$3.0 \times 10^{-7} \text{ kg} \cdot \text{m}^2$
Operating temperature (*)	0...+70°C -40...+85°C
Maximum rotation speed without code failure	6000 rpm
Vibration acceleration (55 to 2000 Hz)	$\leq 100 \text{ m/S}^2$
Maximum shock acceleration	$\leq 2000 \text{ m/S}^2$
Maximum rotation speed	6000 rpm
Protection EN 60 529	IP40
Interface (*)	BiSS C SSI
Type of output code (*)	Gray code Binary code

LIR-DA237T



Absolute rotary encoder

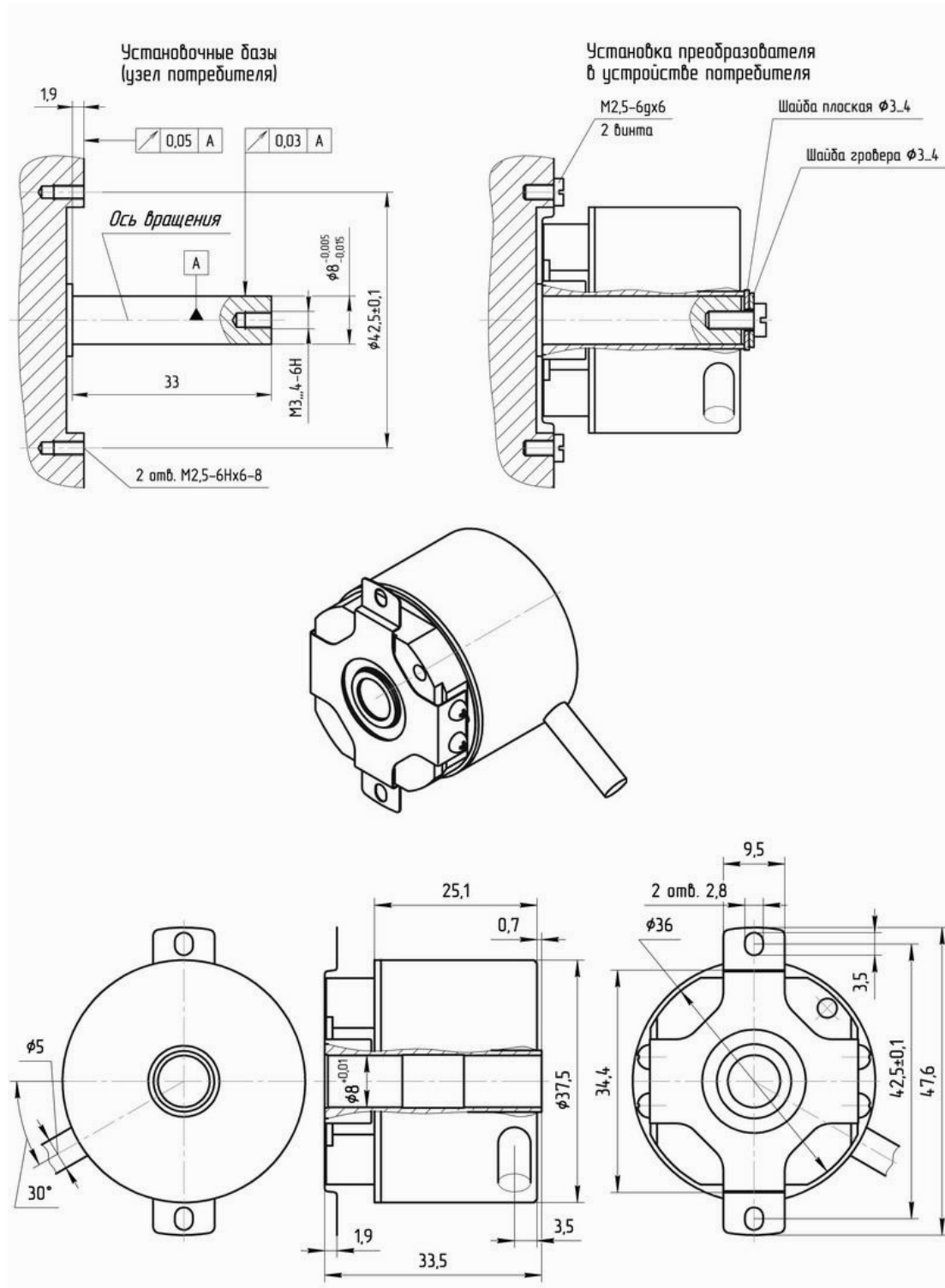
Interface	RS422 (RS422 standard)
Supply voltage	+5 V
Incremental signal 1Vpp 1024 ppr (*) <i>The possibility of issuing an absolute additional incremental sensor signal for positioning at high speeds of rotation of the shaft</i>	No / Yes
Current consumption	≤ 140 mA
Number of digits (*)	8 (256 positions) 9 (512 position) 10 (1024 positions) 11 (2048 positions) 12 (4096 positions) 13 (8192 positions) 14 (16384 positions) 15 (32768 positions) 16 (65536 positions) 17 (131072 positions) 18 (262144 position) 19 (524288 positions) 20 (1048576 positions) 21 (2097152 position) 22 (4194304 positions) 23 (8388608 positions)
Hull option	Cable out radial
Cable length (*)	0.5 meters ; 1 meter ; 2 meters ; 3 meters
Permissible error <i>Defines the absolute error limit at a nominal temperature of 20°C</i>	6 (±30")
Cable connector (*)	RS10TV Male ; RS10TV Female ; DB9 Male ; DB9 Female ; DB15 Male ; DB15 Female ; Flying leads

(*) — The required value is selected when ordering. See order information below.

LIR-DA237T

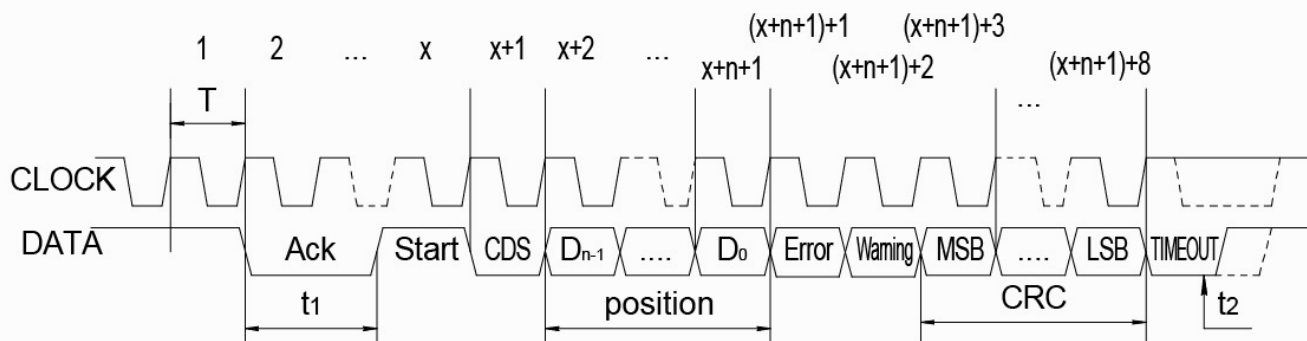


Absolute rotary encoder





Output signals



BISS C

Info $\overline{\text{CLOCK}}$ and $\overline{\text{DATA}}$ are not shown.

CLOCK - input control signal

DATA - output signal

T - not less 100 ns

Position calculation time t_1 :

- unto 19 bit: $\leq 9.5 \mu\text{s}$
- 20 bit: $\leq 9.9 \mu\text{s}$
- 21 bit: $\leq 10.2 \mu\text{s}$
- 22 bit: $\leq 10.6 \mu\text{s}$
- 23 bit: $\leq 11 \mu\text{s}$

Timeout $t_2 \leq 16 \mu\text{s}$ (adaptive for LIR-DA116V, LIR-DA119A.01, LIR-DA120A.02, LIR-DA290A, LIR-DA156A, LIR-DA165A and LIR-DA216 series)

Acknowledgment **Ack** - always set to 0

Bit **Start** - always set to 1

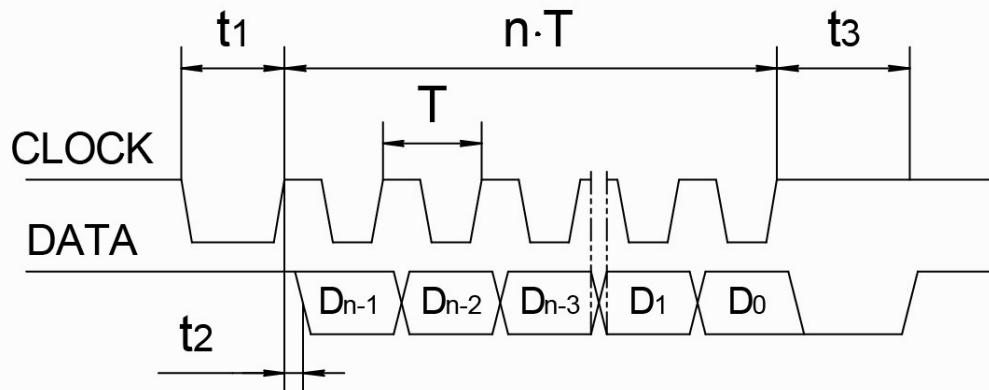
Bit **CDS** - always set to 0

D - position code with **n** bits

Bit **Error** - means error if set to 0

Bit **Warning** - means warning if set to 0

Six bits checksum **CRC**, polinom 0x43, initial value 0



SSI

Info $\overline{\text{CLOCK}}$ and $\overline{\text{DATA}}$ are not shown

CLOCK - input control signal

DATA - output signal with n bits

T - from 0.25 μs to 16 μs

$t_1 \leq 9.5 \mu\text{s}$

$t_2 \leq 0.4 \mu\text{s}$

$t_3 \leq 20 \mu\text{s}$

First CLOCK bus and DATA bus are set to logic 1. After first CLOCK decay DATA bus setting to 0 and encoder is storing position in buffer.

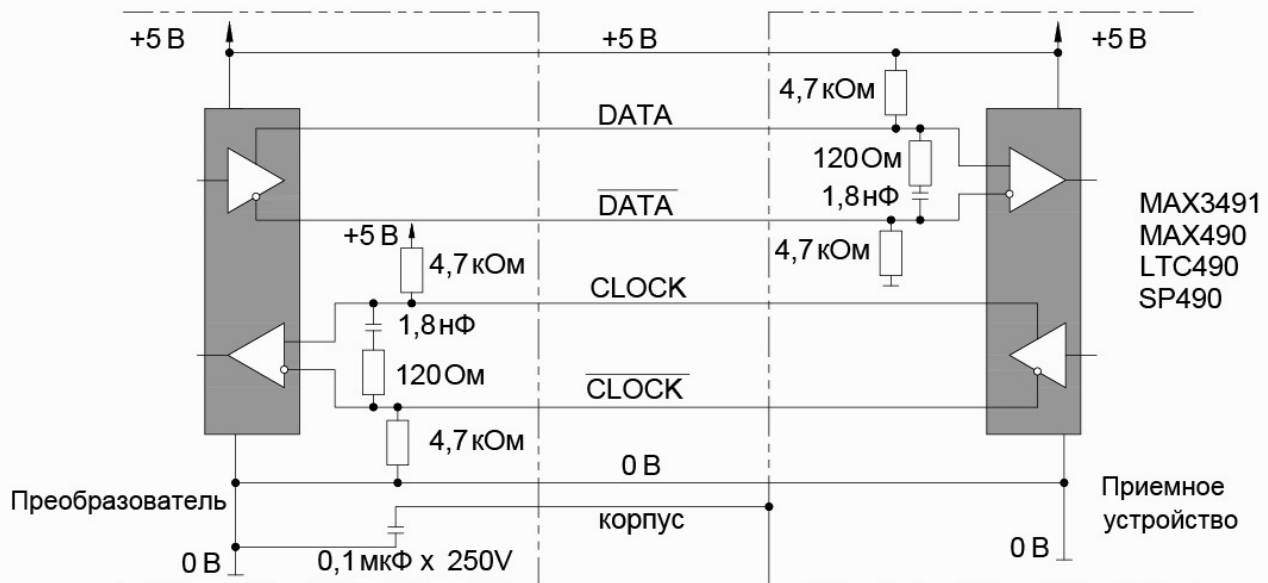
By the CLOCK fronts going bit-by-bit transmission of the fixed code value by DATA bus, then DATA is setting to logic 0 and stays for a time t_3 . At this time, a fixed position code value can be re-read by setting CLOCK logic 0 and sending pulse. Repete may be done an unlimited number of times. After t_3 time DATA setting to logic 1 and encoder is ready for sending new frame. If during code reading state of CLOCK bus is not changing during time more than t_3 , encoder is setting to initial state.

LIR-DA237T

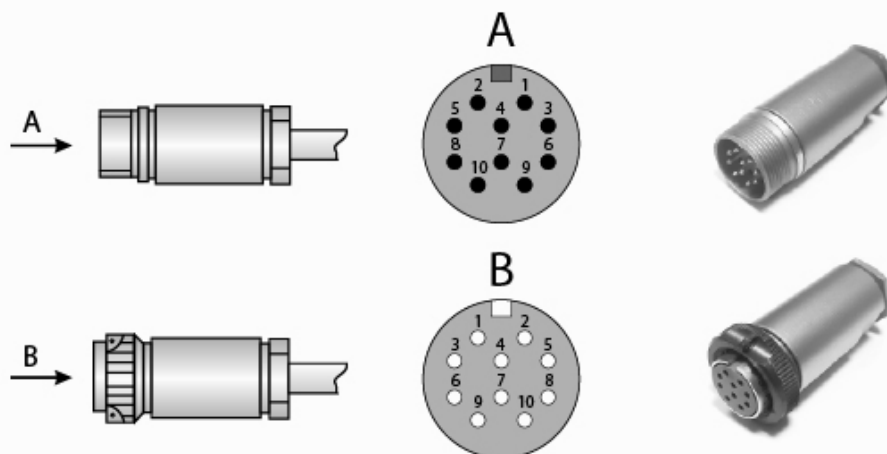


Absolute rotary encoder

Wiring diagram



Connector



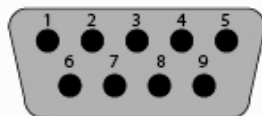
Connector RS10TV. Serial data output:

Assignment	Clock	Data	Clock	Data	—	—	—	Power	0B	Shield
Pin number	5	3	8	6	1	10	7	2	9	4

LIR-DA237T

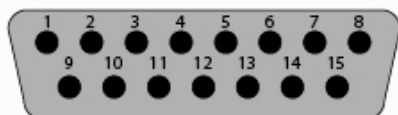


Absolute rotary encoder



Connector DB9. Serial data output:

Assignment	Clock	Data	$\overline{\text{Clock}}$	$\overline{\text{Data}}$	—	—	Power	0B	Shield
Pin number	2	6	3	7	4	8	5	9	1



Connector DB15. Serial data output with incremenatal ~1 Vpp:

Assignment	Data	$\overline{\text{Data}}$	Clock	$\overline{\text{Clock}}$	—	—	A	$\overline{A}$	B	$\overline{B}$	—	—	0V	Power	Shield
Pin number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Flying leads. Serial data output:

Assignment	Clock	Data	$\overline{\text{Clock}}$	$\overline{\text{Data}}$	Power	0V (GND)	Sense Up	Sense 0V
Cable 4 twisted pair	Green	Red	Yellow	Blue	Brown	White	Rose	Grey

Flying leads. Serial with 1 Vpp:

Assignment	Clock	Data	$\overline{\text{Clock}}$	$\overline{\text{Data}}$	A	B	$\overline{A}$	$\overline{B}$	Power	0V	Sense Up	Sense 0V
Cable 6 twisted pair	Green	Red	Yellow	Blue	Violet	Red-Blue	Black	Grey-Rose	Brown	White	Rose	Grey



RS10TV connector

Industrial cable 10-pin connector (plug-socket), which is widely used in many encoders LIR

» [Link to product page](#)



DB9 connector

Cable 9-pin connector D-sub used to connect to the controllers SPPU, some DRO, boards and interface modules

» [Link to product page](#)



Cable for absolute encoders cable

Cable with soldered connectors for connecting absolute encoders LIR

» [Link to product page](#)

Contact information

SKBIS, JSC

Saint-Petersburg, 195009
Kondratevskiy prospekt, 2-A

Phone: **+7(812) 334-17-72**

Fax: **+7(812) 540-29-33**

Email: lir@skbis.ru

SKBIS Center, LLC

Moscow, 109117
Okskaya st, 5/1

Phone: **+7(495) 225-66-16, 709-42-41**

Fax: **+7(495) 225-66-16, #20**

Email: lircenter@skbis.ru